SS-C-1783
March 29, 1973
SUPERSEDING
MIL-C-10316C
September 7, 1972

CLOTH, ASBESTOS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a range of classes and grades of asbestos cloths, having heat and chemical resistance for use as thermal insulation and for other high temperature applications.

1.2 Classification. Asbestos cloth shall be of the following classes and grades, as specified (see 6.2).

Class 1 - Plain (nonmetallic).

Grade AAAA - 99 percent asbestos, by weight, min.
Grade AAA - 95 percent asbestos, by weight, min.
Grade AA - 90 percent asbestos, by weight, min.
Grade U.G. - 80 percent asbestos, by weight, min.
Grade C.G. - 75 percent asbestos, by weight, min.
See industrial designations in table I.

Class 2 - Metallic (wire inserted).
(The same grades of cloths as above shall apply.)
See industrial designations in table II.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-N-281 - Nickel-Copper Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-W-321 - Wire, Copper Alloy.
UU-P-268 - Paper, Kraft, Untreated, Wrapping.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-T-76 - Tape, Pressure-Senstive Adhesive Paper, (For Carton Sealing).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 191/GEN - (General Provisions) Textile Test Methods.
Fed. Test Method Std. No. 191/1400 - Identification of Asbestos.
Fed. Test Method Std. No. 191/4021 - Yarn Number (Linear Density) of Yarn From Package.
Fed. Test Method Std. No. 191/5010 - Determination of Length of Textile Materials.
Fed. Test Method Std. No. 191/5020 - Determination of Width of Textile Materials.
Fed. Test Method Std. No. 191/5030 - Determination of Thickness of Textile Materials.
Fed. Test Method Std. No. 191/5040 - Weight of Cloth Cut, Roll, or Bolt Method.
Fed. Test Method Std. No. 191/5050 - Yarns Per Inch in Woven Cloth.
Fed. Test Method Std. No. 191/5100 - Breaking Strength and Elongation, of Woven Cloth; Grab Method.

FSC 5640

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Class 1, plain nonmetallic. Class 1 asbestos cloth shall be woven of chrysotile asbestos yarn of uniform size and shall contain no rubber or other filling material, except organic fiber.

3.1.2 Class 2, wire-inserted. Class 2 asbestos cloth shall be woven of wire-inserted yarn. The yarn shall be as specified in 3.1.1 and the wire shall be as specified in 3.2.2.

3.2 Composition.

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3.2.1 Asbestos.

3.2.1.1 Hygroscopic moisture. The hygroscopic moisture in the asbestos cloth, based on the weight of cloth without wire, shall average not more than 5 percent when tested as specified in 4.5.1.

3.2.1.2 Asbestos content. The asbestos content in the cloth shall be as listed in 1.2 when tested as specified in 4.5.2.

3.2.1.3 Chemically combined water. The dry asbestos fiber shall contain not less than 12 percent by weight of chemically combined water (water of crystallization) when tested as specified in 4.5.3.

3.2.2 Wire. The wire shall be 0.008 inch plus or minus 0.001 inch in diameter and shall be copper conforming to QQ-W-321 alloy No. 270, or nickel-copper-alloy, class A, conforming to specification QQ-N-281. The wire shall have 1 or 2 plies as specified in table II.

3.3 Color. Unless otherwise specified (see 6.2), the color of the cloth shall be natural.

3.3.1 Color stripe in selvage. A colored cotton yarn stripe shall be woven into each selvage of the cloth, the color shall be in accordance with the grade as follows:

Grade AAAA - No stripe.
Grade AAA - Green.
Grade AA - Red.
Grade U.G. - Blue.
Grade C.G. - Black.

3.4 Physical properties. The physical properties of the cloth shall be in accordance with the requirements as specified in table I or II, as applicable, and as otherwise specified herein.

3.4.1 Weave. Weave shall be in accordance with tables I, and II. Twill weave shall be 2 up and 1 down twill to the right. Broken-twill weave shall be a 1 up and 3 down twill with ends to the right and ends to the left. Otherwise, the twill-weave and the broken-twill weave cloth shall be as specified in table I.

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TABLE I. Physical properties of class 1 (nonmetallic) cloth

Grade (see 1.2 and 6.1)	Weave (see 3.4.1)	Wgt. lbs. per sq. yd.	Thickness (see 3.5.2)	Warp yarn 1/	Filling yarn 1/	Warp ends per inch ±1	Filling picks per inch ±1	Breaking strength as received (min.) Warp	Breaking strength as received (min.) Filling	Industrial designation
U.G.	Plain	1.40 ± .10	.051	16/2/0	16/2/0	19	10	80	40	22P16
U.G.	Plain	1.50 ± .11	.055	10/2/0	10/2/0	9	9	55	50	24P10
U.G.	Plain	1.80 ± .13	.058	14/2/0	14/2/0	20	10	95	40	29P14
U.G.	Plain	1.80 ± .13	.060	10/2/0	10/2/0	16	10	90	30	29P10
U.G.	Plain	2.25 ± .16	.070	10/2/0	10/2/0	18	9	120	50	36P10
U.G.	Plain	2.50 ± .18	.075	10/2/0	10/2/0	20	10	140	55	40P10
U.G.	Broken Twill	2.50 ± .18	.070	14/2/0	14/2/0	27	14	160	70	40H14
U.G.	Double Cloth	4.75 ± .34	.135	10/2/0	10/2/0	37	16	185	80	76P10DC
AA	Plain	1.40 ± .10	.051	16/2/0	16/2/0	19	10	80	40	22P16
AA	Plain	1.80 ± .13	.058	14/2/0	14/2/0	20	10	95	45	29P14
AA	Plain	2.25 ± .16	.070	10/2/0	10/2/0	18	9	115	50	36P10
AA	Plain	2.50 ± .18	.075	10/2/0	10/2/0	20	10	135	55	40P10
AAA	Plain	1.40 ± .10	.051	16/2/0	16/2/0	19	10	85	45	22P16
AAA	Plain	2.10 ± .15	.062	10/2/0	10/2/0	16	9	100	50	34P10
AAA	Plain	2.25 ± .16	.070	10/2/0	10/2/0	18	9	130	55	36P10
AAA	Plain	2.40 ± .17	.065	12/2/0	12/2/0	20	12	105	55	38P12
AAA	Twill	3.00 ± .21	.075	12/2/0	12/2/0	30	10	170	50	48T12
AAAA	Twill	3.33 ± .24	.080	12/2/0	12/2/0	36	12	180	55	53T12
AAAA	Twill	3.70 ± .26	.080	12/2/0	12/2/0	40	14	160	40	59T12

1/ The first figure signifies the number of cut and the second figure signifies the number of plies. Cut of yarn is defined as the number of 100 yard lengths per pound. The tolerance for the single yarn is ± 15 percent.

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TABLE II. Physical properties of class 2 (metallic-wire inserted) cloth

Grade (see 1.2 and 6.1)	Reinforcing wire (3.1.2)	Weave	Wgt. lbs. per sq. yd.	Warp yarn 1/	Filling yarn 1/	Warp ends per inch ±1	Filling picks per inch ±1	Industrial designation
C.G.	Brass	Plain	1.80 ± .13	14/2/1	14/2/1	13	13	30M14
C.G.	Brass	Plain	2.80 ± .20	8/2/2	8/1/1	14	11	45M8
AA	Brass	Plain	3.10 ± .21	10/2/1	10/2/1	19	19	50M10
AA	Brass	Plain	2.75 ± .19	14/2/1	14/2/1	20	10	44M14
AAA	Brass	Plain	2.75 ± .19	8/2/2	8/1/1	14	11	44M8
AAA	Brass	Plain	4.00 ± .28	10/2/2	10/2/2	22	11	64M10
AAA	Brass	Plain	7.00 ± .49	10/2/2	10/2/2	28(7 2/)	12	112M10
AAAA	Monel	Plain	2.75 ± .19	8/2/2	8/1/1	14	11	44M8

1/ The first figure signifies the number of cut, the second figure signifies the number of asbestos yarn plies, and the third figure signifies the number of wire plies. Cut of yarn is defined as the number of 100-yard lengths per pound. The tolerance for the single yarn is ± 15 percent.
2/ Binder warp ends.

3.5 Dimensions.

3.5.1 Width. Unless otherwise specified, (see 6.2) the width of asbestos cloth shall be 40 inches plus or minus 0.500 inch.

3.5.2 Thickness. The thickness of the asbestos cloth shall not vary from the values specified in table I by more than the following tolerances. Commercial thickness and tolerance is acceptable for grade C.G.

Thickness of cloth inches	Permissible Variation in thickness inches
0.050 and under	±0.005
Over 0.050	±0.010-0.005

3.5.3 Length. Unless otherwise specified (see 6.2), the asbestos cloth shall be furnished in rolls, each roll having a length of 50 plus or minus 5 yards. Not less than 90 percent of the number of rolls shall be furnished in one piece, and no roll shall contain more than two pieces, of which the shortest piece shall be not less than 20 yards. The cloth shall be rolled open-width on a convolute or spiralwound chipboard tube. The tube shall have a minimum wall thickness of 0.200 inch with a minimum inside diameter of 2.000 inches. The ends of the tube shall be flush with or extend no more than 1 inch beyond each side of the maximum width of the rolled cloth. Each roll shall be restrained from unwinding by securing with three equally spaced ties of cotton tape. The length shall be tested as specified in 4.5.4.

3.6 Workmanship. The asbestos cloth shall be free of nap, broken or uneven threads, knots, or any other defects or imperfections that may affect the serviceability or appearance of the finished product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.6).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level S-4.

4.3.1.1 Lot. A lot shall consist of all asbestos cloth of one class and grade manufactured as one production run and offered for delivery at one time.

4.3.1.2 Unit of product. The unit of product shall be a single roll of asbestos cloth (see 3.5.3).

4.3.1.3 Cloth samples for physical properties. From each sample roll selected in accordance with 4.3.1, a 1-yard sample of asbestos cloth shall be taken to determine compliance with construction requirements of section 3 and the tests specified herein.

4.3.1.4 Sample specimens. From each sample yard of asbestos cloth selected in accordance with 4.3.1.3, a representative specimen of asbestos yarn, from which the wire has been removed from class 2 cloth, and weighing not less than 5 grams, shall be taken for tests.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in table III. The acceptable quality level (AQL) shall be 2.5 percent defective.

4.3.3 Test samples. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.5.4. Cloth samples selected in accordance with 4.3.1.3 shall be tested as specified in 4.5.5. Sample specimens selected in accordance with 4.3.1.4 shall be tested as specified in 4.5.1 through 4.5.3. Any roll which fails one or more tests or any roll from which a cloth sample or specimen fails one or more tests shall be considered a defective sample roll. AQL shall be 2.5 percent defective.

4.4 Examination. The asbestos cloth shall be examined for the following defects:

TABLE III. Examination for construction

Requirement	Reference	Method
Wire diameter.....	3.2.2	Micrometer
Colored stripe.....	3.3.1	Visual
Weave.....	3.4.1	Visual
Number of yarn plies:		
Warp yarn.....	Table I or II	Visual
Filling yarn.....	Table I or II	Visual
Ply of wire insert:		
Warp yarn.....	Table II	Visual
Filling yarn.....	Table II	Visual
Workmanship.....	3.6	Visual

4.5 Tests methods.

4.5.1 Hygroscopic moisture. Specimens of asbestos yarn selected in accordance with 4.3.1.4 and each weighing not less than 5 grams shall be weighed and then dried for 1 hour at 105° to 110°C. in an electric oven, cooled for 1 hour in a desiccator containing anhydrous calcium chloride and weighed. The loss in weight is termed the hygroscopic moisture. This weight divided by the weight of the specimen and multiplied by 100 gives the percentage of hygroscopic moisture in the specimen. Nonconformance to 3.2.1.1 shall constitute failure of this test.

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4.5.2 Asbestos content. Specimens used in 4.5.1 shall be used to determine the asbestos content. Specimens, each weighing not less than 4 grams, after drying for 1 hour at 105° to 110°C. shall be placed in a desiccator for 1 hour, again weighed, and then placed in a furnace and heated for not less than 1 hour at 800° to 810°C. After removal from the furnace, the specimens shall be cooled for 1 hour in the desiccator, containing anyhydrous calcium chloride, and then again weighed. The weight of the residue shall be divided by 0.860 to determine the original weight of the asbestos content. (The factor 0.860 is based upon an average theoretical value of 14 percent chemically combined water. Chemically combined water subtracted from 100 should yield the proper factor for determination.) The weight of the asbestos content shall be divided by the weight of the dried specimen and the quotient multiplied by 100 to obtain the percentage of asbestos. The average of all determinations shall be taken as the asbestos content. Nonconformance to 3.2.1.2 shall constitute failure of this test.

4.5.3 Chemically combined water. Sample specimens of well-opened asbestos fiber selected in accordance with 4.3.1.4 shall be used to make determinations of chemically combined water. Each sample of asbestos shall weigh approximately 5 grams. The hygroscopic moisture shall be removed as specified in 4.5.1, the weight recorded, and the dried material ignited in an electric furance at 800°C. for 1 hour and the weight again recorded. The percentage of chemically combined water shall be calculated as follows:

$$\text{Percentage of chemically combined water} = \frac{(A-B) \times 100}{A}$$

A = Weight of dried specimen
B = Residual weight of ignited specimen

Nonconformance to 3.2.1.3 shall constitute failure of this test.

4.5.4 Weight and length. Sample rolls of asbestos cloth, selected as specified in 4.3.1, shall be tested for compliance with weight and length requirements in accordance with FED. TEST METHOD STD. NO. 191 as specified in table IV. Nonconformance to the weight requirements of tables I and II or the length requirement of 3.5.3 shall constitute failure of the applicable test.

TABLE IV. Test methods

Test	Specification requirement	Specification test paragraph	Test method
Hygroscopic moisture	3.2.1.1	4.5.1	
Asbestos content	3.2.1.2	4.5.2	1400
Chemically combined water	3.2.1.3	4.5.3	
Weight	Table I or II		5040
Thickness	Table I		5030
Asbestos yarn number:			
Warp yarn	Table I or II		4021
Filling yarn	Table I		4021
Thread count:			
Warp ends	Table I or II		5050
Filling picks	Table I or II		5050
Breaking strength:			
As received:			
Warp	Table I		5100 1/
Filling	Table I		5100 1/
Width	3.5.1		5020
Length	3.5.3		5010

1/ Prior to tests for breaking strength as received, the specimens shall be conditioned to moisture equilibrium in a standard atmosphere having a relative humidity of 50 ± 2 percent and a temperature of 70° ± 2°F. (21° ± 1.1°C.).

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4.5.5 Yarn number, thread count, breaking strength, width, and thickness. Cloth samples selected as specified in 4.3.1.3 shall be tested for compliance with the yarn number, thread count, breaking strength, width, and thickness requirements in accordance with FED. TEST METHOD STD. NO. 191 as specified in table IV. Nonconformance to the applicable requirements of table I, table II, 3.5.1, or 3.5.2 shall constitute failure of the applicable test.

4.4.3 Examination of preparation for delivery. An examination shall be made for the requirements listed in table V. to determine that markings, materials, workmanship and contents comply with the requirements of section 5. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. The closed shipping containers shall be examined for closure defects. The inspection level shall be S-2 in accordance with MIL-STD-105 and the AQL shall be 4.0 expressed as defects per 100 units.

TABLE V. Preparation for delivery defects

Examine	Defects
Marking (interior and exterior container)	Incorrect; incomplete; illegible; omitted; of improper size, location sequence or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose or inadequate sealing, strapping or stapling when required.
Contents	Number of rolls per shipping container more or less than required.
Containers	Not as specified, not close-fitting for level A or B.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Asbestos cloth in the lengths and widths as specified (see 6.2) shall be put-up in rolls. Each roll shall be completely wrapped in kraft paper, grade B, 60-pound basis weight (24 x 36 -500) conforming to UU-P-268.

5.1.2 Level B. Each asbestos cloth roll shall be wrapped in kraft paper, 30 pound basis weight conforming to UU-P-268.

5.1.3 Level C. Each asbestos cloth roll shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Asbestos cloth of like class, grade, thickness, width and length, in quantities as specified (see 6.2), packaged as specified in 5.1 shall be packed in an overseas type close-fitting shipping container conforming to PPP-B-601, PPP-B-621, PPP-B-636 or PPP-B-640, at the option of the contractor. The nailed wood box and the cleated plywood box, if selected shall have case liners conforming to MIL-L-10547 or a heat sealed plastic bag conforming to L-P-378, 3 mil thick. Case liners may be omitted from the fiberboard boxes, provided all center, edge seams, manufacturers points and staples are covered by weather-resistant tape conforming to PPP-T-76. The boxes shall be closed and strapped in accordance with the appendix to the applicable container specification, level A requirements.

5.2.2 Level B. Each roll of asbestos cloth packaged as specified in 5.1.2 shall be packed in a close fitting fiberboard box conforming to PPP-B-636, class domestic. The closure shall be in accordance with the Method I requirements for domestic boxes as specified in the appendix to the box specification.

5.3 Marking. In addition to any other markings specified in the procurement document, the asbestos cloth rolls shall be marked in accordance with Fed. Std. No. 123 for civil agencies, or MIL-STD-129 for military agencies, as applicable (see 6.2).

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6. NOTES

6.1 Intended use. Asbestos cloth is intended for use as thermal insulation and for other high-temperature applications.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class, grade, and weight of cloth required (see 1.2 and 6.1).
(c) Quantity in linear yards required.
(d) Color required if other than natural (see 3.3).
(e) Width if other than that specified (see 3.5.1).
(f) Length if other than that specified (see 3.5.3).
(g) Level of packaging and level of packing required (see 5.1 and 5.2).
(h) Whether palletization is required.
(i) Whether special marking is required (see 5.3).

MILITARY INTERESTS:

Custodians:

Army - ME
Air Force - 84

Review Activity:

Army - MI

User Activity:

Navy - YD

Military Coordinating Activity:

Army - ME

Project No. 5640-0265

Civil Agency Coordinating Activities:

GSA-FSS

Preparing Activity:

GSA-FSS

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